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High Performance Processors and Systems

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Name
Last Name

Problem 1 (50%)	
Problem 2 (50%)	
Total (100%)	

Problem 1

1. To implement cache coherency one of the protocols is the MESI concurrency protocol implemented for a write-invalidate write-back cache. Describe briefly the algorithm (also in term of graph) and complete the following table.

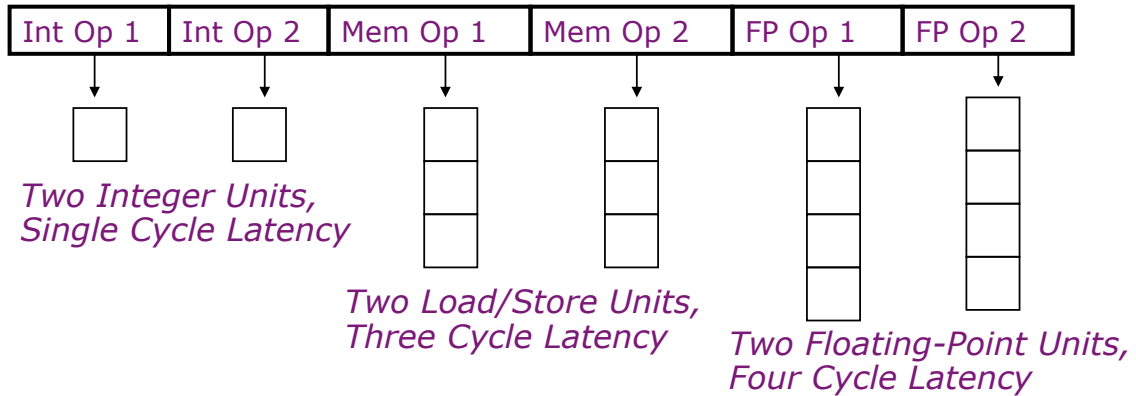
State	Cache up to date?	Memory up to date?	Others have a copy?	Cache can respond other's reads?
Modified				
Exclusive				
Shared				
Invalid				

2. Consider the following access pattern on a two-processor system with a direct-mapped, write-back cache with one cache block and a two cache block memory. Assume the MESI protocol is used, with write-back caches, write-allocate, and invalidation of other caches on write (instead of updating the value in the other caches).

Time	After Operation	P1 cache block state	P2 cache block state	Memory at block 0 up to date?	Memory at block 1 up to date?
0	P1: read block 1	Exclusive (1)	Invalid (0)	Yes	Yes
1	P2: read block 0				
2	P1: write block 0				
3	P2: read block 2				
4	P1: read block 1				
5	P2: write block 1				
6	P1: read block 1				
7	P2: write block 1				
8	P1: read block 0				
9	P2: write block 1				
10	P2: write block 1				
11	P1: read block 1				
11	P2: write block 1				
12	P1: read block 1				
13	P2: read block 1				

Problem 2

Considering the following VLIW "architecture":



Considering the following portion of assembly, describe the corresponding VLIW code:

```
Loop:    LD F0, 0(R1)
         LD F6, 8(R1)
         LD F10, 16(R1)
         LD F14, 24(R1)
         FADD F4, F0, F2
         FADD F8, F6, F0
         FADD F12, F10, F14
         FADD F16, F14, F2
         SD F4, 0(R1)
         SD F8, 8(R1)
         ADD R1, 32
         SD F12, -16(R1)
         BNE R1 R2 Loop
         SD F16, -8(R1)
```

How many FLOPS/cycle?

Answer P2

Int1

Int 2

M1

M2

FP+

FPx

[illegible]